Ammonium Powered Aircraft Flight Dynamics and Controls

Project Lead:	Dr. Rob Hewson
Academic Supervisor:	Prof. Rafael Palacios
Department:	Department of Aeronautics
Course:	MEng Aeronautical Engineering
Module:	AERO60004 Group Design Project
Academic Year:	2022/23
Student:	Kee Kwan Liu
CID:	01847234
Group:	APR05
Date	19/6/2023

Abstract

This report focuses on the stability analysis of an innovative ammonium-powered aircraft and provides a comprehensive analysis of its stability characteristics. The study includes the methodology for estimating stability derivatives, making corrections for individual components, and investigating both the dynamical and static stability aspects of the aircraft. The primary objective of this research is to evaluate the feasibility of using ammonium as a fuel source for aircraft propulsion by assessing stability and the ability to meet CS-25 requirements. The analysis concludes that the aircraft exhibits static and dynamic stability.

Acknowledgement

The author wishes to extend his sincere appreciation and gratitude to Professor Palacios for his invaluable guidance and unwavering support throughout the project. Professor Palacios' expertise, dedication, and numerous hours of consultation have been instrumental in shaping the outcome of this research endeavor. Special recognition should be given to the APR team for their outstanding design efforts throughout recent weeks and their consistent delivery of top-notch results and data.

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Nomenclature

Acronyms

ax	Approximation approach
CG	Center of Gravity

Greek Symbols

α	Angle of attack
β	Sideslip angle
δa	Deflection angle of aileron
δe	Deflection angle of elevator
δr	Deflection angle of rudder
δ	Control surface angle of deflection
ϵ_{d_α}	Downwash gradient
ϵ_{s_β}	Sidewash gradient
Γ	Dihedral
λ	Eigenvalue
ω_d	Damping frequency
ω_n	natural frequency frequency
ω_p	Propeller angular velocity
ϕ	Bank angle
σ	Damping rate
σ_p	Propeller blade area to disc area
ζ	Damping ratio

Roman Symbols

AR	Wing aspect ratio
$\bar{c}$	Mean aerodynamic chord of wing
θ	Pitch angle
$\tilde{\phi}$	Fluid velocity potential
A	State matrix
a	Wing lift curve slope
a_h	Horizontal tail lift curve slope
B	Control matrix
b	Wing span
c	Mean aerodynamic chord of wing
C_D	Coefficient of drag
C_L	Coefficient of lift
C_w	Coefficient of weight
CAP	Control anticipation parameter
d_p	Diameter of propeller
e	Oswald efficiency
g	Gravitational acceleration
I	Identity matrix
I_{xx}	Roll second moment of inertia
I_{yy}	Pitching second moment of inertia
I_{zz}	Yawing second moment of inertia
J	Propeller advance ratio
K_n	Static margin
l	Rolling moment
l_f	Length of fuselage
M	Pitching moment
m	Aircraft mass
n	Yawing moment
q	Pitch rate
S	Wing planform area
S_f	Fuselage maximum cross sectional area
S_h	Horizontal planform area
S_v	Vertical tail planform area
U	Velocity along x wind axis
U_e	Equilibrium velocity
U_∞	Free stream velocity
W	Velocity along z wing axis
W	Weight
W_r	Reachability matrix
X	Force along x wind axis
Y	Sideforce
Z	Force along z wind axis

1 Introduction

Flight dynamics and controls are vital for ensuring the safe operation of aircraft. By comprehending the aircraft's behavior in flight, engineers can develop systems that effectively maintain stability and appropriately respond to disturbances. The design of an aircraft involves an iterative process, where the controls team plays a crucial role in validating and verifying the design parameters. This requires close collaboration and expertise from multiple subteams, such as aerodynamics, structures, and propulsion, who contribute their specialized knowledge to the overall design process. Together, these teams work towards creating a well-balanced and efficient aircraft that meets the required safety and performance standards.

Although conventional aircraft and ammonia-powered aircraft employ similar techniques for analysing stability, the use of ammonia as fuel introduces unique considerations that impact aerodynamics, structural design, and propulsion configuration. The need to liquefy ammonia and address structural concerns results in unconventional design elements, such as a thicker airfoil and a larger tail, which can contribute to different stability characteristics.

2 Stability Estimates Methodology

For the purpose of preliminary analysis, several main assumptions are assumed to reduce complexity and focus on the key aspect of interests, enabling a more efficient and feasible analysis process.

Aircraft stability analysis is a complex multidisciplinary task which incorporates aerodynamics, structures, propulsion, systems, and ground operation. The interplay and integration of these disciplines necessitate an iterative approach until a stable and acceptable design is redefined, ensuring all design and regulations requirements are met. Figure 1 illustrates the analysis flowchart.

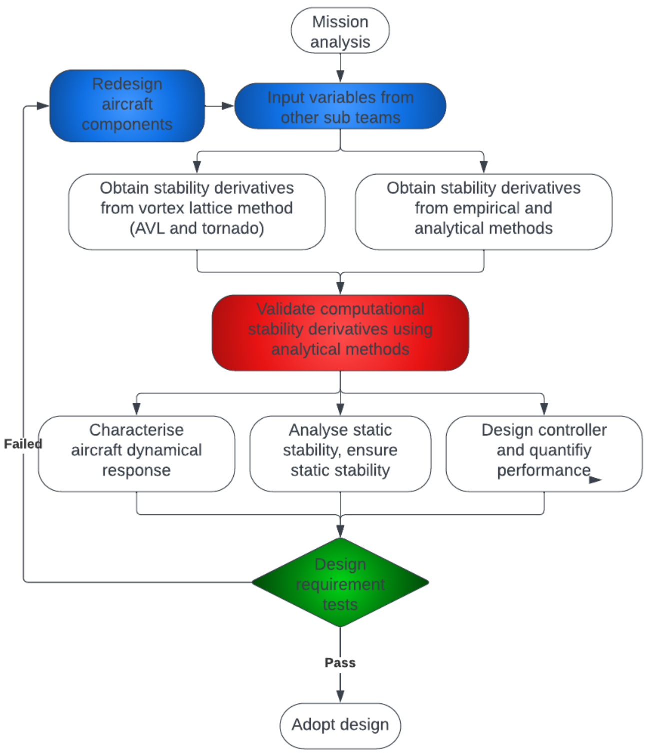

Figure 1: Stability analysis flow chart

2.1 Vortex Lattice Method

Computational methods for estimating derivatives are important in aircraft design and analysis due to their efficiency, accuracy, and ability to handle complex systems. These methods enable a rapid assessment of stability derivatives, speeding up the iterative design process. In parallel with computational methods, analytical and empirical methods provides a means of verifying the accuracy and reliability of the computational models.

For this particular project, Vortex Lattice Method implemented in Tornado and AVL is chosen due its simplicity and computational efficiency as compared to alternatives e.g. CFD or panel method. VLM functions via discretization of 3D bodies into lattice of horseshoe vortices[1]. Boundary conditions are enforced at the collocation point at 3/4 chord of all panels to generate a matrix of influence coefficients.

$$\sum_{j}^{N} A_{ij,vlm}\Gamma_{j,vlm} = -U_{\infty,vlm}\sin(\theta_{vlm}) \tag{1}$$

2.2 Stability Estimates Criticism

Both analytical methods and VLM software such as Tornado and AVL assumes an ideal flow[2], neglecting all viscous effects for simplicity. Small perturbation is assumed which may not apply in certain scenario. 3D flow physics are also neglected which in reality contributes to span-wise effects. While aeroelasticity and slosh dynamics are present, both methods assumes rigid body. In VLM software, propellers, fuselage, wing box, and friction drag are assumed to be negligible with wing, tail and control surfaces being the main contributor of stability derivatives.

2.3 Longitudinal Derivatives

Longitudinal stability derivatives are aerodynamic parameters that quantify the stability characteristics of an aircraft in the longitudinal motion. The estimates[3] provided are originally expressed in dimensional form; however, they can be converted into non-dimensional form. Table 1 presents a comparison between the estimate and AVL approach of longitudinal stability derivatives. Although most of the derivatives show a general consensus, there are notable differences observed in C_{M_U} and C_{M_W}.

	AVL[7]	Estimates	Deviation
C_{X_U}	−0.1829	−0.1068	78.96%
C_{Z_U}	−1.5618	−1.2502	62.30%
C_{M_U}	1.7812	0.0000	100.00%
C_{X_W}	0.4870	0.3282	32.40%
C_{Z_W}	−5.8519	−5.8852	−0.30%
C_{M_W}	1.5027	−0.1750	111.65%
C_{X_q}	1.5330	−0.7000	145.66%
C_{Z_q}	−6.8843	−4.3354	37.02%
C_{M_q}	−20.1366	−20.7765	−3.18%

Table 1: Longitudinal Stability Derivatives Comparison

2.3.1 U-Derivatives

With chain rule, the state of interest can be expressed in terms of the individual effects of the total freestream velocity and angle of attack, with U_∞ being the most significant.

$$\frac{\partial \square}{\partial U} = \frac{\partial \square}{\partial \alpha}\frac{\partial \alpha}{\partial U} + \frac{\partial \square}{\partial U_\infty}\frac{\partial U_\infty}{\partial U} \approx \frac{\partial \square}{\partial U_\infty} \tag{2}$$

The follow expressions are used to estimate the U-derivatives:

$$X_U = -\rho U_\infty S C_D \tag{3}$$

$$Z_U = -\rho U_\infty S C_L \tag{4}$$

Due to our mission profile at $M = 0.45$, compressibility effects are small thus $M_U \approx 0$. Due to the heavy dependence of X_U and Z_U on lift and drag, the AVL values are believed to be more accurate and are used for these parameters. Given that M_U primarily encompasses compressibility effects, it is highly probable that AVL has significantly overestimated these effects, leading to the use of estimated values instead.

2.3.2 W-Derivatives

Similar to U-derivatives, W-derivatives are expressed in terms of the individual effects of α and U_∞, with effects due to α being the main contributor.

$$X_W = -0.5\rho U_\infty S(\frac{2a}{\pi AR e} + C_L) \tag{5}$$

$$Z_W = -0.5\rho U_\infty S(a + C_L) \tag{6}$$

$$M_W = -0.5\rho U_\infty S\bar{c}K_n a \tag{7}$$

For the same reason as U-derivatives, AVL values of X_W and Z_W are used to better approximation of lift and drag, estimates value of M_W are adopted due compressibility effects overestimation.

2.3.3 Pitch Rate q-Derivatives

The q-derivatives indicate the effect of a pitch rate about the aircraft center of gravity. A major contribution of these derivatives comes from the horizontal tailplane due to its aft position relative to the CG.

$$X_q = 0.5\rho U_\infty S_h(x_{ac_h} - x_{cg})[\frac{2C_{L_h}a_h}{\pi e AR_h} + CL_h] \tag{8}$$

$$Z_q = -0.5\rho U_\infty S_h(x_{ac_h} - x_{cg})(a_h + C_{D_h}) \tag{9}$$

$$M_q = (x_{ac_h} - x_{cg})\frac{\partial Z}{\partial q} - (z_{ac} - z_{cg_h})\frac{\partial X}{\partial q} \tag{10}$$

When a positive pitch rate occurs, the horizontal tailplane generates a negative X force, resulting in stabilizing effects and a negative value for X_q. Hence, it is believed that AVL has substantially overestimated the influence of the wing and vertical tail, resulting in a positive X_q. Consequently, the estimated value for X_q was utilized instead.

2.3.4 Criticism of Longitudinal Derivatives

The aforementioned estimations are based on a stick-fixed case, neglecting the inclusion of the elevator and trim-tab states[3]. In order to consider a stick-free case, it becomes necessary to account for the states of δE elevator deflection and δT trim-tab deflection. To reduce complexity, only the most significant contributors, such as the effects of α and U_∞, are taken into consideration. However, it is important to acknowledge that this approach may overlook smaller effects. Similar to the assumptions made in the Vortex Lattice Method (VLM), a small angle approximation[4] is employed, and the analysis is most suitable near the equilibrium point, with the neglect of aeroelasticity[5].

2.4 Lateral Derivatives

The estimation of lateral stability derivatives in this study incorporates a combination of two methods: Roskam's statistical-based approach[6] and Babista's first-principle formulation[8]. Roskam's method offers reasonable approximations, but it is primarily derived from statistical data and is interpolated using conventional turbojet aircraft with a certain degree of sweep. In contrast, Babista's approach formulates derivative estimates based on first-principle principles involving integration along the wing span. Babista's approach is used when Roskam's estimates show high statistical dependencies on sweep, which is not applicable to our design with zero sweep. Both methodologies are employed to ensure a comprehensive analysis of the aircraft's lateral dynamics. The estimates are presented in non-dimensional form, and the conversion to dimensional values is outlined in Appendix.

	Beech 99[6]	Tornado[7]	Estimates	Deviation
C_{Y_β}	-0.59	-0.553	-0.9902	78.96%
C_{Y_p}	-0.19	0.0709	0.1153	62.62%
C_{Y_r}	0.39	0.4408	0.8005	81.60%
C_{n_β}	0.08	-0.2522	0.2863	213.52%
C_{n_p}	0.019	0.0517	0.0219	57.64%
C_{n_r}	-0.20	-0.2548	-0.2909	14.17%
C_{l_β}	-0.13	0.0959	-0.1494	255.79%
C_{l_p}	-0.50	-0.5909	-0.4235	-28.33%
C_{l_r}	0.14	0.1355	0.0752	-44.5%

Table 2: Lateral Stability Derivatives Comparison

2.4.1 Roll Rate p-Derivatives

The determination of these derivatives relies heavily on the wing, which contributes significantly to the roll damping. By employing strip theory[8], the lift and drag components that induce roll and yaw moments are analyzed, leading to the formulation of the equations presented below.

$$C_{Y_p} = 2C_{Y_{\beta_v}}(z_v cos\alpha - l_v sin\alpha)/b \tag{11}$$

where $C_{Y_{\beta_v}}$ is estimated via

$$C_{Y_{\beta_v}} = -K_v a_v(1 + \frac{d\sigma}{d\beta})\eta_v\frac{S_v}{S} \tag{12}$$

$$C_{n_p} = -\frac{2}{Sb^2}\int_0^s \left(C_L - \frac{\mathrm{d}C_D}{\mathrm{d}\alpha}\right) cy^2 \,\mathrm{d}y \tag{13}$$

$$C_{l_p} = -\frac{2}{Sb^2}\int_0^s (C_D + a)\, cy^2 \,\mathrm{d}y \tag{14}$$

Due to high dependency on the wing in C_{l_p} and C_{n_p} and the lack of influence from fuselage, VLM is theoretically more accurate and is adopted in this report. It is noteworthy that VLM is likely to have a more reliable estimation of C_{Y_p} since it is typically considered negligible[8] unless the aircraft has a high degree of sweep.

2.4.2 Yaw Rate r-Derivatives

Yaw rate derivatives describe the changes in the aircraft's yaw rate in response to various inputs. These derivatives are mainly contributed by the wing and vertical tail.

$$C_{Y_r} = -2C_{Y_{\beta_v}}(l_v cos\alpha + z_v sin\alpha)/b \tag{15}$$

$$C_{n_r} = \{\frac{C_{n_r}}{C_L^2}C_{L_w}^2 + \frac{C_{n_r}}{C_{D_o}^2}C_{D_o}^2\}_w - \{\frac{C_{Y_r}}{b}\}_v \tag{16}$$

$$C_{l_r} = \frac{4}{Sb^2}\int_0^s (C_L)\,cy^2\,\mathrm{d}y \quad (17)$$

Table 2 shows that the derivatives are largely in agreement with each other due to both approaches correctly identifies the wing and the tail as the main contributor. As a result, VLM values are used for yaw rate derivatives.

2.4.3 Sideslip β-Derivatives

Considered one of the most important lateral derivatives due to its connection with lateral general stability criterion which will be further explained in section 5. β-derivatives are greatly influenced by the wing and tail[9] with non-negligible effects by the fuselage as explained in next section.

$$C_{Y_\beta} = -\eta_v \frac{S_v}{S} a_v (1-\epsilon_{s_\beta}) \quad (18)$$

$$C_{n_\beta} = \eta_v \frac{S_v l_v}{Sb}(1-\epsilon_{s_\beta}) a_v \quad (19)$$

$$C_{l_\beta} = \{-\eta_v \frac{S_v z_v}{Sb} a_v (1-\epsilon_{s_\beta})\}_w + \{-0.08\eta_v(1-\epsilon_{s_\beta})a_h \frac{S_v b_h}{Sb}\}_h \quad (20)$$

Due to the absence of fuselage effects in VLM, it is considered less accurate compared to empirical estimates. As a result, the estimated values for C_{n_β} and C_{l_β} are considered more accurate and are therefore utilized.

2.4.4 Criticism of Lateral Derivatives

Relative to longitudinal derivatives, the fuselage and propellers exert a more significant influence on lateral derivatives. As the aforementioned estimates for lateral derivatives consider only the wing and tail, corrections from various individual components will have to be considered. Strip theory assumes a two-dimensional, incompressible flow with thin airfoil sections.
However, the unconventional aspects of the aircraft, such as its high taper ratio and thick airfoil, may deviate from the methodologies employed and require additional considerations.

2.5 Control derivatives

Control derivatives play a vital role in defining the relationship between control inputs and the dynamic behavior of the aircraft. For the purpose of preliminary analyse, they can be estimated using Saudrey's[10],[11] approach. These derivatives are utilized in the development of a state-space model and control system for the aircraft, enabling the modelling and analysis of control surface effects within the model.

3 Effects and Corrections

Corrections are necessary in the estimation of aircraft derivatives to account for the effects of other components that significantly influence the dynamics of the aircraft. While the primary focus may be on the wing and tail, it is important to recognize that various other components, such as the fuselage and propellers, can have a notable impact on the overall aircraft behavior. These components introduce additional aerodynamic forces and moments that need to be considered for accurate derivative estimations.

3.1 Sidewash Gradient

The lateral stability of an aircraft can be significantly influenced by the sidewash[9] induced on the vertical tail due to the presence of wingtip vortices generated by the main wing. The impact of the sidewash gradient on the roll and yaw stability can be mathematically represented by equations (18) and (19). The sidewash gradient acts as a connecting link between the wingtip vortices and the lateral stability of the aircraft, highlighting the complex aerodynamic interplay involved in lateral stability analysis. Equation (21) shows the modelling of sidewash gradient at vertical tail:

$$\epsilon_{s_\beta} \equiv \frac{\partial \epsilon_s}{\partial \beta} = -\frac{\kappa_v \kappa_\beta}{\kappa_b}\frac{C_L}{AR} \quad (21)$$

The sidewash gradient depends on the ratio of wingtip vortex strength to that generated by an elliptic wing with equivalent lift coefficient and aspect ratio κ_v and the vortex span factor κ_b. Both κ_v and κ_b are determined analytically from the solution of Prandtl's lifting-line equation[12]. Equation of κ_β can be found in Appendix.

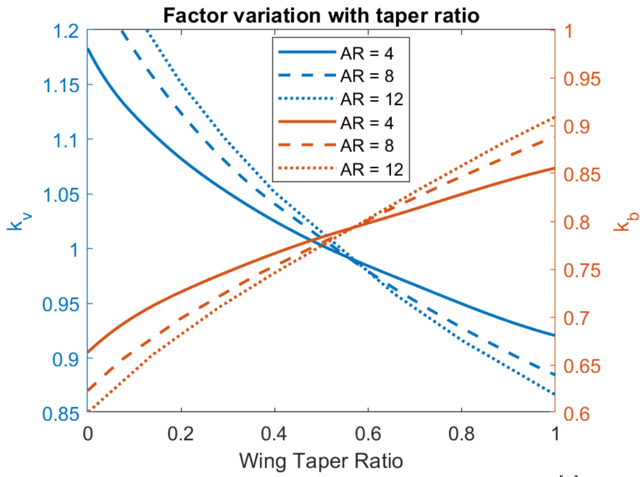

Figure 2: Factor variation with taper ratio[9]

$$\kappa_b = \frac{\frac{\pi}{4} + \sum_{n=2}^{\infty} \frac{nA_n}{(n^2-1)A_1}\cos(n\pi/2)}{1+\sum_{n=2}^{\infty}\frac{A_n}{A_1}\sin(n\pi/2)} \quad (22)$$

	κ_v	κ_b	κ_β	ϵ_{s_β}
Parameters	0.92	0.86	0.56	−0.0376

Table 3: Sidewash gradient parameters

3.2 Downwash Gradient

The downwash[9] induced by the wing on the tailplane has a notable impact on the trim and static stability of the aircraft, leading to a reduction in the effective angle of attack and, consequently, decreasing its stabilizing effectiveness. Achieving accurate results necessitates precise wind tunnel data or highly accurate computational simulations. In cases where such data is limited, analytical estimation methods are employed to approximate the effects of downwash.

$$\epsilon_{d_\alpha} \equiv \frac{\partial \epsilon_d}{\partial \alpha} = \frac{\kappa_v \kappa_p}{\kappa_b}\frac{C_L}{AR} \quad (23)$$

	κ_p	ϵ_{s_β}
Parameters	0.43	0.23

Table 4: Downwash gradient parameters

3.3 Dihedral Effects

Dihedral is frequently utilized to improve roll stability by taking advantage of the effects of sideslip. When the aircraft experiences sideslip, the windward half of the wings experiences an increased angle of attack, while the lee-side half experiences a decreased angle of attack. This lift differential creates a rolling moment that stabilises the aircraft. While dihedral improves roll stability, an excessive amount of stability can compromise roll performance by limiting the aircraft's ability to bank and execute turns effectively. This phenomenon is notably prominent due to the inherent dihedral effect of high wing configuration, which amplifies the dihedral effects[12].

Using small angle approximation, the expression for the effects on roll stability is given by:[9]

$$(C_{l_\beta})_\Gamma = -\frac{2\sin\Gamma}{Sb}\int_0^{b/2} ayc\ \mathrm{d}y = -0.0235 \tag{24}$$

3.4 Fuselage Effects

The fuselage's impact on static stability is generally destabilizing due to the intricate aerodynamic forces and moments it generates. Empirically, these effects are represented or approximated using the following models:[9]

$$\Delta C_{m_{\alpha_f}} = \frac{2S_f l_f}{S\bar{c}}[1 - 1.76\frac{d_f}{c_f}^{1.5}] \tag{25}$$

Similar to the destabilising effects on pitch stability, effects on yawing moment:

$$\Delta C_{n_{\beta_f}} = \Delta C_{m_{\alpha_f}}\frac{\bar{c}}{b} = \frac{2S_f l_f}{Sb}[1 - 1.76\frac{d_f}{c_f}^{1.5}] \tag{26}$$

	$\Delta C_{m_{\alpha_f}}$	$\Delta C_{n_{\beta_f}}$
Deviation	-0.170	-0.017

Table 5: Fuselage effects

3.4.1 Criticism of Fuselage Effects

In initial or preliminary design stages, the effect of the fuselage on airplane lift is often neglected as a first approximation. This simplification allows for a quick and rough estimate of the lift distribution. However, for more accurate and detailed analyses, it is necessary to consider the aerodynamic moments generated by the pressure forces acting on the fuselage. To account for these effects, advanced methods such as panel methods, viscous computational fluid dynamics, and wind tunnel tests should be utilized.

3.5 Propeller effects

The effects of a spinning propeller on the trim and static stability of an aircraft [9]can be divided into two primary categories: direct effects and indirect effects. The direct effects result from the aerodynamic forces and moments exerted directly on the propeller. In contrast, the indirect effects stem from the interaction between the propeller's slipstream and other surfaces of the aircraft. The influence of the propeller slipstream on various surfaces of an airplane, such as the wing, tail, and fuselage, is highly intricate and cannot be precisely analyzed through traditional analytical methods.

3.5.1 Propeller Direct Effects

When a propeller rotates and moves forward through the air at an angle to the freestream, it can generate not only thrust T but also a side force Y and a yawing moment n. A propeller has the capability to influence both the trim and static stability of an airplane. The thrust, force and yaw moment coefficients[13] can computed via (27) - (29).

$$T = \frac{\sigma_p q_p \mathcal{R}}{2}\left(\frac{\pi}{J}\right)\left(\frac{2\pi}{3J}\bar{C}_l - C_d\right) \tag{27}$$

$$C_{N_{\alpha,p}} = \frac{\sigma_p}{2}\left\{\bar{C}_l + \frac{aJ}{2\pi}\ln\left[1 + \left(\frac{\pi}{J}\right)^2\right] + \frac{\pi}{J}C_d\right\} \tag{28}$$

$$C_{n_{\alpha,p}} = \frac{-\sigma_p}{2}\{\frac{2\pi}{3J}\bar{C}_l + \frac{a}{2}\left[1 - \left(\frac{J}{\pi}\right)^2\ln\left(1 + \left[\frac{\pi}{J}\right]^2\right)\right] - \frac{\pi}{J}C_d\} \tag{29}$$

Equation (27) gives $\bar{C}_l$ of the propeller. Combined with J the propeller advance ratio from equation (30), the force and yaw coefficient can be computed.

$$J \equiv \frac{2\pi V_\infty}{\omega_p d_p} \tag{30}$$

Figure 3 presents the derivatives in ratio form[13] as a function of advance ratio.

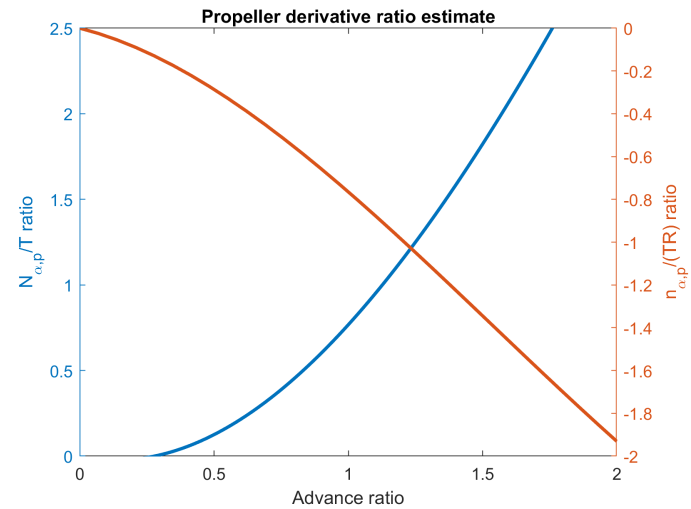

Figure 3: Propeller derivative ratio

The propeller's yawing moment and side force are typically quantified using dimensionless aerodynamic coefficients. The yawing moment coefficient demonstrates a linear relationship with the propeller's angle of attack α_p, while the side force coefficient exhibits a linear relationship with the propeller's sideslip angle β_p, as showin in (31) and (32).

$$C_{n_p} = C_{n_{\alpha,p}}\alpha_p \tag{31}$$

$$C_{Y_p} = C_{Y_{\beta,p}}\beta_p = -C_{N_{\alpha,p}}\beta_p \tag{32}$$

The propeller induces a change in the yawing moment in the lateral direction, which can be expressed as follows:[9]

$$\Delta(C_n)_p \approx \Delta(C_{n_\alpha})_p\alpha_p + \Delta(C_{n_\alpha})_p\alpha_p \tag{33}$$

where

$$\Delta(C_{n_\alpha})_p = \frac{2d_p^3}{S_w b_w}(1 - \epsilon_{d_\alpha})\frac{C_{n_{\alpha,p}}}{J^2} \tag{34}$$

$$\Delta(C_{n_\beta})_p = \frac{2d_p^2 l_p}{S_w b_w}(1-\epsilon_{s_\beta})\frac{C_{N_\alpha,p}}{J^2} \tag{35}$$

Figure 4 illustrates the variation of the propeller correction on C_{n_β} with fixed thrust as the design parameters of angular velocity and propeller diameter are modified.

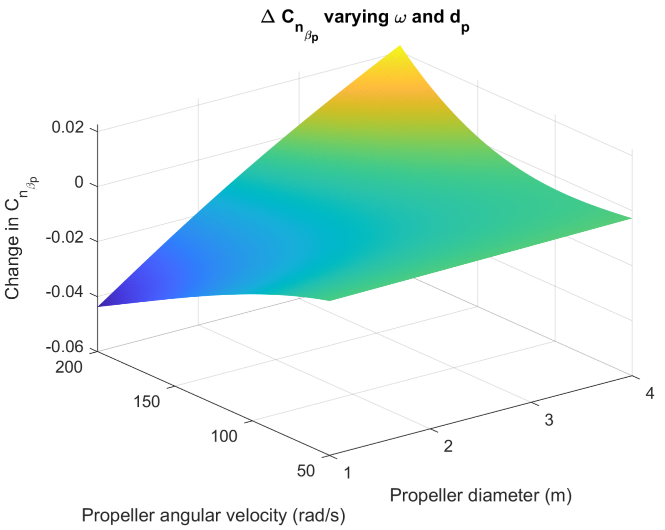

Figure 4: $\Delta(C_{n_\beta})_p$ variation with ω and d_p

Longitudinally, the propeller's contribution to the overall airplane pitching moment coefficient can be expressed as follows:

$$\Delta(C_m)_p \approx \Delta(C_{m_\alpha})_p \alpha_p \tag{36}$$

where

$$\Delta(C_{m_\alpha})_p = \frac{2d_p^2 l_p}{S_w b_w}(1-\epsilon_{d_\alpha})\frac{C_{N_\alpha,p}}{J^2} \tag{37}$$

At different flight conditions, Table 6 displays the alterations in stability derivatives for this aircraft caused by the presence of propeller.

	Take off	Cruise
$\Delta(C_{n_\alpha})_p$	-0.0010	-0.0017
$\Delta(C_{n_\beta})_p$	-0.0128	-0.0164
$\Delta(C_{m_\alpha})_p$	-0.0094	-0.0131

Table 6: Propeller effects on stability derivatives

3.5.2 Criticism of Propeller Effects

The method employed simplifies the problem by neglecting the propeller-induced velocity effect and assuming a propeller blade with constant chord. This approach overlooks the change in direction of sectional lift force. The helical path of induced velocities in the slipstream and its influence on the rest of the aircraft are not taken into account.

To attain a more precise understanding of the propeller's influence, the analysis can be conducted using blade element theory or propeller vortex theory, as mentioned in the references[14]. This method discretise the vortices to create a vortex sheet that represents the flow leaving each propeller blade. However, considering the current preliminary design stage, a simplified analytical method is preferred over the high-fidelity approach.

4 State Space Model

A state space model is a mathematical representation[15] that captures the dynamic behavior of a system using state variables, inputs, and outputs. In the preliminary design stage of an aircraft, the development of a state space model enables engineers to gain a thorough understanding of the aircraft's characteristics, facilitating informed decision-making throughout the design process.

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Ix + Ou \end{aligned} \tag{38}$$

In order to simplify the analysis, it is assumed that all states are directly observable. It disregards the presence of noise and disturbances typically encountered in real-life situations, resulting in a deterministic system without stochasticity. According to the Curie principle, which states that no asymmetric effect can arise from a symmetric cause, two separate state models are derived: one for the longitudinal dynamics and another for the lateral dynamics.

4.1 Criticism of State Space Models

The application of the state space method is primarily suitable for analyzing equilibrium flight conditions due to linearisation. However, regions near the boundaries of the flight envelope necessitate accurate flight simulation data. Additionally, constant thrust and drag forces[3] are assumed, with the thrust aligned along the x-axis. Traditionally, the analysis is divided into longitudinal and lateral motion, with the coupling between them often overlooked. This oversight was evident in the case of the German prototype He-162 in 1944[16], which experienced structural failure during high-speed roll maneuvers that could not be explained without considering the influence of longitudinal-lateral coupling[9]. Moreover, time-delaying and nonlinear effects, such as yaw divergence during roll maneuvers or aerodynamic yaw departure at high angles of attack are neglected[9].

4.2 Linearised Longitudinal System

The longitudinal state space model would incorporate input states of U, W, q, and θ and control state of δe in (39)[3]. The state matrix and control matrix are given by (40) and (41).

$$x_{\text{long}} = [U, W, q, \theta]^T \quad \text{and} \quad u_{\text{long}} = [\delta_e] \tag{39}$$

$$A_{\text{long}} = \begin{bmatrix} \frac{1}{m}\frac{\partial X}{\partial u} & \frac{1}{m}\frac{\partial X}{\partial w} & \frac{1}{m}\frac{\partial X}{\partial q} & -gcos(\theta) \\ \frac{1}{m}\frac{\partial Z}{\partial u} & \frac{1}{m}\frac{\partial Z}{\partial w} & (u_0 + \frac{1}{m}\frac{\partial Z}{\partial q}) & -gsin(\theta) \\ \frac{1}{I_{yy}}\frac{\partial M}{\partial u} & \frac{1}{I_{yy}}\frac{\partial M}{\partial w} & \frac{1}{I_{yy}}\frac{\partial M}{\partial q} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{40}$$

$$B_{\text{long}} = \begin{bmatrix} \frac{1}{m}\frac{\partial X}{\partial \delta_e}, \frac{1}{m}\frac{\partial Z}{\partial \delta_e}, \frac{1}{I_{yy}}\frac{\partial M}{\partial \delta_e}, 0 \end{bmatrix}^T \tag{41}$$

The stick-fixed system consists of three degrees of freedom, namely axial, normal, and pitch. When analyzing the linearized longitudinal motion of a typical airplane, the eigenvalues associated with this system exhibit two complex pairs. These complex pairs represent two distinct oscillatory modes: the Short Period Phugoid Oscillation (SPPO) mode and the Phugoid mode.

4.3 Linearised Lateral System

The lateral dynamics of an aircraft when disturbed from equilibrium flight involve a intricate interplay between sideslip, roll, and yaw. The linearized representation of this lateral stick-free system is expressed as[17]:

$$x_{\text{lat}} = [\beta, p, r, \phi]^T \quad \text{and} \quad u_{\text{lat}} = [\delta_a, \delta_r]^T \tag{42}$$

$$A_{\text{lat}} = \begin{bmatrix} \frac{Y_\beta}{u_0} & \frac{Y_p}{u_0} & \frac{Y_r}{u_0} - 1 & \frac{g\cos\theta_0}{u_0} \\ L_\beta & L_p & L_r & 0 \\ N_\beta & N_p & N_r & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{43}$$

$$B_{\text{lat}} = \begin{bmatrix} \frac{Y_{\delta_a}}{u_0} & L_{\delta_a} & N_{\delta_a} & 0 \\ \frac{Y_{\delta_r}}{u_0} & L_{\delta_r} & N_{\delta_r} & 0 \end{bmatrix}^T \tag{44}$$

The lateral system gives rise to three distinct modes: Dutch roll, spiral, and roll subsidence mode. These modes will be discussed in detail in a subsequent section.

4.4 Dynamical Stability

In the preliminary design stage, comprehending and analyzing the aircraft's dynamical response holds paramount importance. It offers valuable insights into stability and control characteristics while significantly influencing performance and maneuverability. Additionally, compliance with regulatory requirements[18], notably CS Part 25, assumes a pivotal role in ensuring adherence to safety standards and performance criteria for commercial operation eligibility. The expressions below[8] gives the characteristic of a dynamic mode for conjugate pairs.

$$\lambda_{1,2} = -\sigma \pm \omega_d i = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \tag{45}$$

where the damping ratio is given by

$$\zeta = -\frac{\lambda_1 + \lambda_2}{2\sqrt{\lambda_1\lambda_2}} = \frac{\sigma}{\sqrt{\sigma^2 + \omega_d^2}} \tag{46}$$

where the natural frequency is given by

$$\omega_n = \sqrt{\lambda_1\lambda_2} = \sqrt{\sigma^2 + \omega_d^2} \tag{47}$$

with the half-amplitude time and period of oscillation T given by

$$t_{\frac{1}{2}} = ln(2)/\sigma \quad \text{and} \quad T = 2\pi/\omega_n \tag{48}$$

4.4.1 Methodology

Initially, an exploration of theoretical approximate solutions was undertaken to gain engineering insights into the dynamical modes of the aircraft. This investigation aimed to understand how the positioning and parameters of the wing and tailplane would influence the aircraft's dynamical behavior. While this approach provided a rapid comprehension of variations, such as the impact of increasing the area under lift distribution in relation to the spanwise position on the damping rate of roll subsidence, it did not yield meaningful values for all metrics, as elaborated in the subsequent subsection. To address this limitation, dynamic mode metrics are calculated using the aforementioned two-state space model.

Through collaboration with the tailplane designer and the baseline configuration team, it was determined that the aircraft's dynamical stability was nearing a neutrally stable state and its modes did not exhibit satisfactory handling qualities. To address this issue, a sequential approach was adopted. Initially, the location and geometry of the tailplane were temporarily fixed. Subsequently, the aerodynamics team implemented design modifications that accounted for both dynamical stability and aerodynamic considerations. This iterative process involved fine-tuning the sizing of the tailplane to enhance the overall stability of the aircraft. If the achieved results did not meet the desired criteria, the process would iterate, initiating another cycle of analysis and adjustments.

4.4.2 Longitudinal Approximation and State Space Solution

The approximate solution of SPPO and Phugoid mode are expressed as follows.[8]

$$\tau\lambda_{\text{SPPO}} = \frac{-a_h S_h}{2S}\frac{ml_h^2}{I_{yy}} \pm i\sqrt{-\frac{2m^2\bar{c}}{\rho_e S I_y y}\frac{\partial C_m}{\partial\alpha} - \frac{1}{4}\left(\frac{a_h S_h}{S}\frac{ml_h^2}{I_{yy}}\right)^2} \tag{49}$$

$$T_{\text{SPPO}} = 2\pi\left[\frac{-\rho_e U_e S\bar{c}}{2I_{yy}}\frac{\partial C_m}{\partial\alpha} - \frac{\rho_e^2 U_e S_h^2 a_h l_h^4}{16I_{yy}^2}\right]^{-\frac{1}{2}} \tag{50}$$

$$\lambda_{\text{Phugoid}} = 0 \pm i\sqrt{2}\frac{C_L}{\tau} \quad \text{and} \quad T_{\text{Phugoid}} = 2\pi\frac{U_e}{g} \tag{51}$$

where the τ the non-dimensional factor is given by

$$\tau = \frac{m}{0.5\rho_e U_e S} \tag{52}$$

Although the initial approximations of the Short Period Pitch Oscillation (SPPO) and Phugoid modes provide valuable engineering intuition, computational values are considered more accurate. The SPPO approximation focuses solely on pitching moment terms, disregarding downwash lag, while the phugoid approximation neglects drag and compressibility effects and assumes a zero pitching angle. Despite these limitations, the approximation methods have proven useful in initial wing and tail plane design.

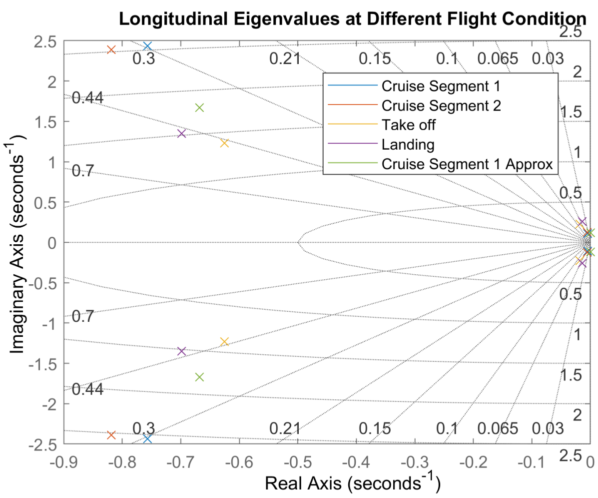

Figure 5: Longitudinal Eigenvalues

	Phugoid	SPPO	Phugoid$_{ax}$	SPPO$_{ax}$
λ	$-0.0043 \pm 0.1086i$	$-0.7574 \pm 2.4378i$	$0.000 \pm 0.1212i$	$-0.6682 \pm 1.6727i$
ω_d	0.1086	2.4378	0.1212	1.6727
ω_n	0.1086	2.5527	0.1212	1.8012
ζ	0.0393	0.2967	0.0000	0.3710
$t_{\frac{1}{2}}$	162.2366	0.9151	∞	1.0374
T	57.8305	2.4614	51.8566	3.4884

Table 7: Longitudinal modes of first cruise segment

While all four flight conditions are analysed, only the metrics of first cruise segment is shown here due to page limit. First segment is arguably more important due to a higher altitude which reduces the effectiveness of the surfaces with decreasing density.

Table 7 presents the longitudinal dynamical modes obtained through both the approximation and state space approaches. Although the two approaches exhibit similar values and trends, the approximation method predicts a higher damped SPPO mode and a lighter damped phugoid mode. These discrepancies can be attributed to the neglect of certain terms in the approximation method. Among the two modes, the phugoid approximation yields a value closer to the computational result than SPPO, likely due to the simplicity of the expression that correctly considers C_L as a key component. However, considering the underlying assumptions, the state space solution is considered to be more accurate.

In Figure 5, the longitudinal eigenvalues are depicted for various flight conditions. It can be observed that the takeoff and landing conditions exhibit a lower natural frequency for SPPO compared to the cruise condition due to lower velocity. The eigenvalues for the cruise segments are relatively close to each other due to their proximity in flight conditions. It is important to note that neglected ground effects are likely to have an impact on the aircraft's dynamics, potentially resulting in different eigenvalues.

4.4.3 Lateral approximation and computational solution

The approximation method is not adopted for Dutch roll due to its complex nature and the interdependence of sideslip, bank, and yaw, which limits its ability to provide meaningful engineering intuition. However, the approximate solutions for roll subsidence and spiral modes can be expressed as follows.[8]

$$\lambda_{\text{roll}} = C_{l_p}\frac{mb^2}{\tau I_{xx}} + C_{n_r}\frac{mb^2}{\tau I_{zz}} \tag{53}$$

$$\lambda_{\text{spiral}} = \frac{C_L}{\tau}\frac{C_{l_\beta}C_{n_r} - C_{l_r}C_{n_\beta}}{C_{l_\beta}C_{n_p} - C_{l_p}C_{n_\beta}} \tag{54}$$

The approximation methods used are considered less accurate compared to computational methods, similar to the longitudinal solutions. Roll subsidence approximations only take into account the wing while neglecting other components. In both approximations, terms that are considered "small" are assumed to be negligible. As a result, computational metrics are adopted as the final results.

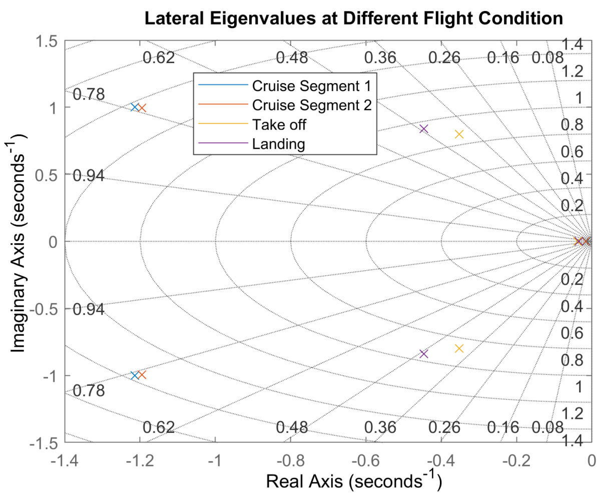

Figure 6: Lateral Eigenvalues

Table 8 illustrates the lateral modes of the aircraft. The approximate values are found to be rather accurate, with deviation within 5%. In both approaches, the spiral mode is very close to the imaginary axis, meaning it will require some pilot or autopilot control input to trim the aircraft. Similar trends as the longitudinal system can be found in figure 6, a Dutch roll mode with lower natural frequency can be observed, indicating a slower oscillation of Dutch roll.

	Dutch Roll	Roll	Spiral	Roll_{ax}	$\text{Spiral}_{\text{ax}}$
λ	$-1.2142 \pm 1.0010i$	-8.0889	-0.0147	-4.9216	-0.0180
ω_d	1.0010	0.0000	0.0000	0.0000	0.0000
ω_n	1.5736	8.0889	0.0147	4.9216	0.0180
ζ	0.7716	1.0000	1.0000	1.0000	1.0000
$t_{\frac{1}{2}}$	0.5709	0.1408	47.1569	0.0857	38.4983
T	3.9928	∞	∞	∞	∞

Table 8: Lateral modes of first cruise segment

4.4.4 Stability Diagrams

Stability diagrams provide a visual representation of how variations in a parameter influence the stability of an aircraft. They offer a concise and effective means of studying the dependence of aircraft stability on specific parameters.

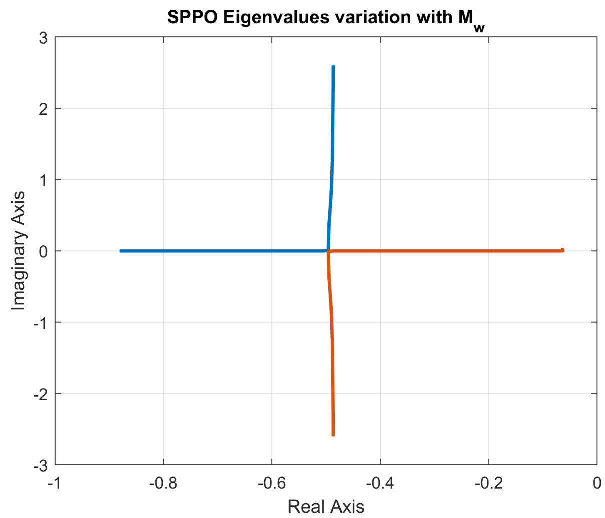

Figure 7: SPPO Eigenvalues variation with M_w

Figure 7 plots λ_{SPPO} by varying M_w from -5e-2 to 1e4. Since the static margin K_n is negatively proportional to M_W, the plot illustrates that as the static margin transitions from positive to negative, the roots of the system's characteristic equation shift from complex to real and quickly become positive. This reveals that the stability of the SPPO mode is significantly influenced by the static margin, and when the static margin is statically unstable, the corresponding dynamical mode also becomes unstable.
From the observed trends in Figure 8, it can be concluded that an increase in the yaw stability coefficient (C_{n_β}) results in a more negative damping rate, indicating improved spiral stability. Conversely, a decrease in the roll stability coefficient ($-C_{l_\beta}$) leads to similar effects, but it also diminishes roll stability, which is an undesirable outcome. The cut-off at the top signifies the spiral stability boundary where the spiral mode becomes unstable, indicating that the design meets the criterion for spiral stability. Another important note is that even if the design falls in the region of spiral divergence, the damping rate would not be high. It is worth noting that even if the design falls within the region of spiral divergence, the damping rate remains relatively low. In such cases, the aircraft can be stabilized effectively through pilot or autopilot

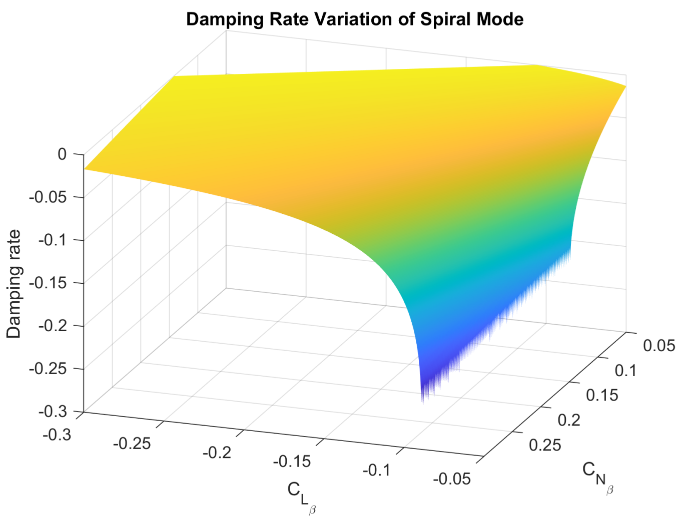

Figure 8: Damping rate variation with spiral motion

inputs.

4.5 Controllability analysis

Controllability analysis is crucial in a state space as it allows engineers to assess the ability to control the system's states as well as the development of controllers.

$$W_r \triangleq \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \dots & \mathbf{A}^{k-1}\mathbf{B} \end{bmatrix} \tag{55}$$

The reachability matrix for both the longitudinal and lateral systems has a theoretical rank of n, indicating reachability[19]. However, physical limitations, like control surface deflection constraints, make the pure mathematical conclusion inaccurate. While the Controllability Gramian Matrix is commonly utilized in state space analysis, it is primarily beneficial for high-dimensional problems involving millions of dimensions, such as in fluid dynamics to understand the degrees of controllability[20] of a few particular directions. In flight dynamics, where there are only four states, Controllability Gramian Matrix is not particularly helpful.

4.6 Open Loop response

Special emphasis is placed on analyzing the transient response of the system. As there is greater interest in the transition behaviour rather than reference tracking ability of open loop system, the impulse rather than step response of both system are analysed to gain insight in the dynamics.

4.6.1 Longitudinal Open Loop Response

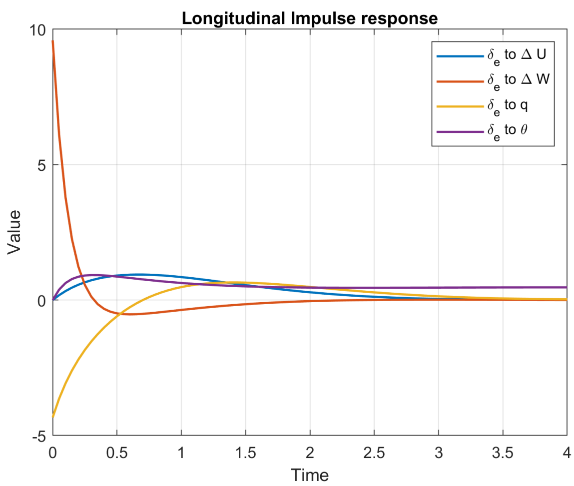

Figure 9: Longitudinal Open Loop Response

All states apart from θ are able to return to its initial position within 3 seconds. θ represents an angular displacement state, and without a feedback system, it lacks the inherent ability to self-correct. Shortly after the disturbance, the pitch rate initially shows a sharp spike and gradually dampens down, indicating a stable static stability. Closely linked to the eigenvalues of a mode, the impulse response of a system reflects the influence of its eigenvalues on its behavior following a sudden input or disturbance. As such, oscillatory behaviour is exhibited on all states due to both the oscillatory nature of SPPO and phgugoid mode.

4.6.2 Lateral Open Loop Response

Similar to the longitudinal system, all states are able to restore itself with ϕ and β settling in a different location, indicating a stable lateral system. It is also observed that aileron input excites significantly more rolling moment, signifying its effectiveness in roll control than the rudder. Despite the presence of two non-oscillatory modes, the coupling between different states is likely to induce the oscillatory behaviour as shown in the impulse response.

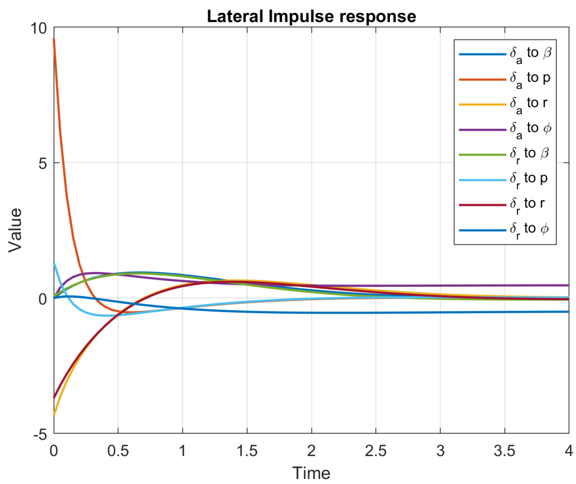

Figure 10: Lateral Open Loop Response

4.7 Frequency response

The frequency response analysis serves as a valuable method for assessing the handling qualities[22] of an aircraft in the frequency domain. The magnitude response at a specific input frequency indicates the sensitivity of the system's states. When the pilot's input extends beyond the bandwidth in an aggressive control, typically defined by the -3dB cutoff frequency, the aircraft may exhibit inadequate response to the pilot's commands, resulting in significant handling concerns.

4.7.1 Longitudinal Frequency Response

The dynamic modes of an aircraft can be associated with the break frequencies observed in the Bode diagram.

The frequency response of velocity U is dominated by the phugoid mode, characterized by a high gain at zero frequency that sharply decreases after the SPPO break frequency around 2.6e0. This aligns well with the physical interpretation of longitudinal modes, where the phugoid mode typically exhibits low-frequency, high-amplitude oscillations, while velocity perturbations in the SPPO mode are usually insignificant. The response of the θ parameter follows a similar trend as the velocity U response, driven by the same underlying reasons.

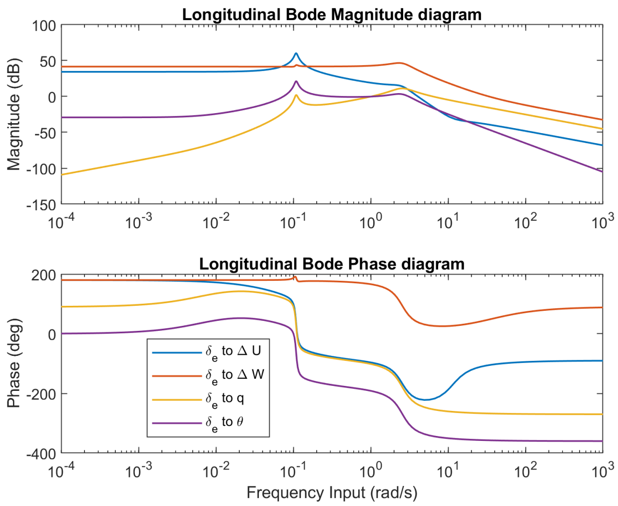

Figure 11: Longitudinal Bode Diagram

As the frequency input increases, all responses significantly decrease below the bandwidth, indicating challenging handling qualities during high-intensity maneuvers. However, within the range of frequencies relevant to human operation, the longitudinal system demonstrates satisfactory sensitivity, particularly in terms of the U and W parameters.

4.7.2 Lateral Frequency Response

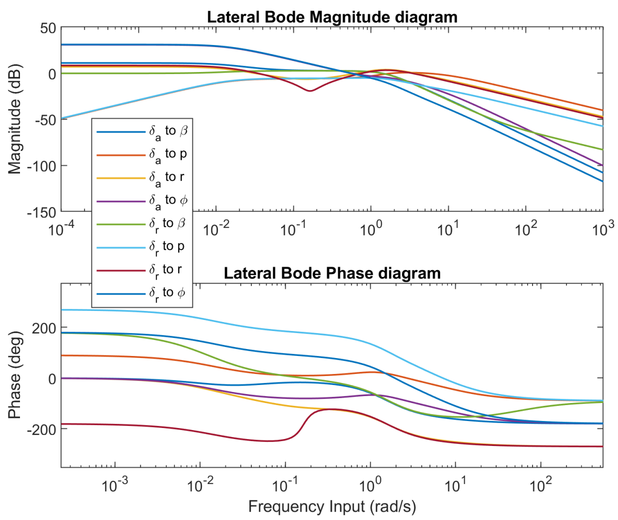

Figure 12: Lateral Bode Diagram

The aircraft's yaw response to rudder input exhibits a gain ratio of approximately 28.6 dB with negligible phase lag at a frequency of approximately 1.5e-2, the spiral mode break frequency. This indicates the aircraft's capability to track the pilot's input, allowing for easy correction of the aircraft's path during spiral mode.

The response of roll rate p to rudder input in the frequency domain reveals a cross-coupling situation. At zero frequency, the gain approaches negative infinity decibels ($-\infty$ dB), indicating that no significant roll rate response is anticipated. Between the spiral and Dutch roll break frequency, around 1.6e0, both the gain and phase remain relatively constant, exhibiting a characteristic frequency response behavior associated with classical Dutch roll dynamics[21].

As the frequency increases to the roll subsidence break frequency approximately 0.8e1, most state responses drops below the bandwidth frequency, sensitivity of the states from aileron and rudder input decreases. The system will attenuate any pilot input, indicating a much more difficult handling issue than spiral mode.

The Dutch roll peak from the β response due to aileron input can be clearly shown with a gain of 1.3 dB, this means that the pilot would experience a slight oscillatory sideslip behaviour.

4.8 Dynamic Handling quality

The pilot's perspective on aircraft control is essential when assessing the handling quality. In addition to examining the frequency response, there are various methods, such as the Cooper-Harper rating system[22], which assigns levels from 1 to 4 to characterize the handling quality, with level 1 representing the highest quality. In this section, a category B flight phase is used as reference.

Since human passengers are sensitive to acceleration, the acceleration sensitivity and control anticipation parameter CAP is more important than the orientation or velocity and is given by[9]:

$$\text{acceleration sensitivity} \equiv \frac{\partial n_{\text{load}}}{\partial \alpha} = \frac{1}{W}\frac{\partial L}{\partial \alpha} = \frac{1}{C_W}\frac{\partial C_L}{\partial \alpha} \tag{56}$$

$$\text{CAP} \equiv (\omega_n^2)_{SPPO} / \frac{\partial n_{\text{load}}}{\partial \alpha} = \frac{(\omega_n^2)_{SPPO}}{L_\alpha / W} = \frac{(\omega_n^2)_{SPPO}}{a/C_W} \tag{57}$$

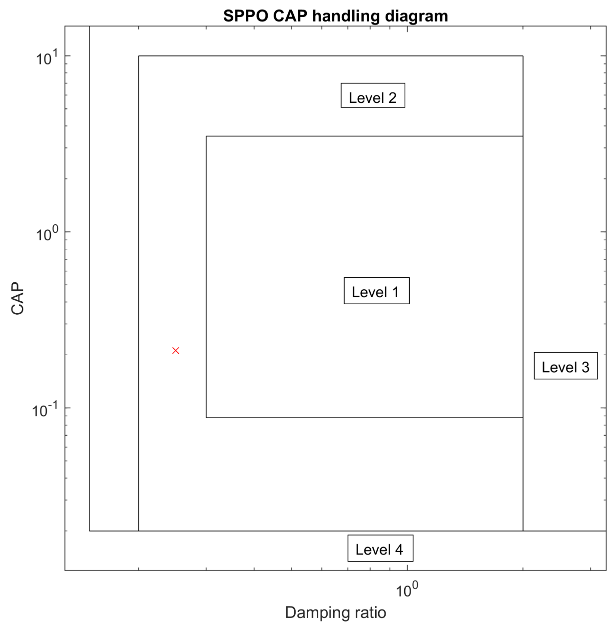

Figure 13: SPPO CAP Diagram

Figure 13 demonstrates that the damping ratio of the SPPO mode does not meet the criteria for level 1 handling quality. From figure 7, adjusting the static margin can shift the damping ratio but would impact the longitudinal static stability, introducing a trade-off. Similar criteria exist for other modes, such as the phugoid mode's ζrequirement, the roll subsidence maximum time constant requirement, and the Dutch roll mode's damping ratio, ζ, $\zeta\omega_n$, and ω_n requirements. Appendix provides a detailed description of these mode requirements. Table 9 provides an overview of the mentioned dynamic modes.

	Quality Level	Description
SPPO	Level 2	Acceptable
Phugoid	Level 2	Acceptable
Roll Subsidence	Level 1	Satisfactory
Dutch Roll	Level 1	Satisfactory
Spiral	Level 1	Satisfactory

Table 9: Handling Quality of Dynamical Modes Summary

4.9 Flight Simulation Comparison

Validating the state space model with a flight simulator is crucial help assess the model's predictive capability and enhances confidence in its ability to accurately represent the aircraft's performance.

Mode	State Space	Flight Simulation[23]
SPPO	$-0.7574 \pm 2.4378i$	$-1.2679 \pm 2.5772i$
Phugoid	$-0.0043 \pm 0.1086i$	$-0.0088 \pm 0.1035i$
Dutch Roll	$-1.2142 \pm 1.0010i$	$-0.2185 \pm 1.2381i$
Roll Subsidence	-8.0889	-1.3964
Spiral	-0.0147	0.0180

Table 10: Flight Simulation and State Space Comparison

Longitudinally, the eigenvalue values of SPPO and the phugoid mode display similar outcomes. However, in the case of SPPO, there is a discrepancy of 67% in σ and 5.7% in ω_d. This difference can be attributed to the complexity of pilot input and the coupling of the phugoid mode, resulting in a higher damping ratio and natural frequency. The mismatch between the models is likely influenced by the purely mathematical nature of the state space model compared to Xplane, which relies on the geometry file. Additionally, due to the pilot releasing control after the initial artificial disturbance, the oscillation component remains largely unaffected, leading to a much closer damping frequency. In contrast, the phugoid mode exhibits a 104% difference in σ, indicating a more damped mode. This can be attributed to the sensitivity of low magnitude σ to effects that are of similar order but neglected in the state space model, such as increased drag experienced in flight simulation.

In terms of lateral dynamics, the flight simulator data suggests a lighter damped Dutch roll mode. This could be attributed to the coupling between roll subsidence and spiral modes, as well as imprecision in input and calculations due to the involvement of sideslip, yaw, and bank angle. Similarly, the roll subsidence mode displays a lower eigenvalue due to coupling effects. The perturbation used to induce roll subsidence in the flight simulator is likely to deviate from assumed small perturbation. Additionally, it is possible that the simulator employs a different model with larger differences in fidelity compared to the longitudinal system. The flight simulator exhibits a diverging spiral mode, whereas the state space model predicts a converging spiral mode. Possible explanations for this discrepancy include the coupling of dihedral effects, fin effects, and the high sensitivity to imprecision given the very low magnitude of the spiral mode.

5 Lateral Static Stability

Lateral stability is crucial for aircraft as it dictates the stability, control, and trimming of the aircraft's lateral motion.

5.1 General Lateral Stability Criterion

The requirement for lateral static stability necessitates that the aircraft generates a yawing or rolling moment to return to its equilibrium state when subjected to a minor disturbance. This leads to the general criterion[24] for lateral static stability:

$$\frac{\partial C_n}{\partial \beta} \equiv C_{n_\beta} > 0 \quad \text{and} \quad \frac{\partial C_l}{\partial \beta} \equiv C_{l_\beta} < 0 \tag{58}$$

Table 11 shows a comparison with values of some competitors. Given the huge vertical tail design, a greater yaw stability is anticipated. s a medium-sized aircraft, the design exhibits slightly higher roll stability compared to the first two competitors, but lower roll stability than a Boeing 747.

	Beech 99[6]	KingAir200[25]	Boeing747[25]	APR
C_{n_β}	0.08	0.12	0.18	0.2863
C_{l_β}	-0.13	-0.13	-0.28	-0.1494

Table 11: Comparison of lateral stability quality

5.2 Decision of C_{l_β} and C_{n_β}

Typically, a desirable lateral handling quality is achieved with a C_{n_β} ranging from 0.06 to 0.15 per radian[9]. However, due to the unique design requirements of an ammonium-powered aircraft and the unconventional wing configuration, a larger vertical tail was incorporated, leading to larger C_{n_β}. Considering the aerodynamic, structural, and stability considerations, a compromise was reached with a C_{n_β} value of 0.2863, providing acceptable handling quality with an increased level of yaw stability.

Likewise, an optimal C_{l_β} value typically ranges from 0 to -0.1 per radian. An excessive roll stability could hinder pilot control, and would be detrimental to the aircraft. Considering aerodynamic and control concerns, the horizontal tail area (S_h) needs to be sufficiently large. Additionally, the damping rate of the spiral mode, as depicted in figure 8, imposes a constraint on the lower limit of C_{l_β}. Consequently, a value of C_{l_β} = -0.1494 was selected to address these concerns.

5.3 Lateral Trim Analysis

A trim analysis is performed to evaluate the necessary control deflection for achieving equilibrium in an aircraft. By balancing the sideforce, roll moment and yaw moment resulting, a general equation in matrix form can be obtained[26].

$$\begin{bmatrix} C_{Y,\beta} & C_{Y,\delta_a} & C_{Y,\delta_r} \\ C_{l,\beta} & C_{l,\delta_a} & C_{l,\delta_r} \\ C_{n,\beta} & C_{n,\delta_a} & C_{n,\delta_r} \end{bmatrix} \begin{bmatrix} \beta \\ \delta_a \\ \delta_r \end{bmatrix} = - \begin{bmatrix} C_w \phi \\ (\Delta C_l)_p \\ (\Delta C_n)_p \end{bmatrix} \tag{59}$$

The lateral trim analysis takes the first cruise segment condition as the equilibrium point. The change in $(C_l)_p$ is approximated to be zero due to the propeller configuration and symmetry, inducing no rolling moment. The change in $(C_n)_p$ is found using equation (33).

The general equation derived from balanced lateral equations of motion is useful to determine the capability to trim, since the required deflection for banking is under the strict regulation. By varying the banking angle, the trim deflection plot is shown in Figure (14).
The deflection angles for sideslip, aileron input, and rudder input show a linear relationship with the bank angle. The

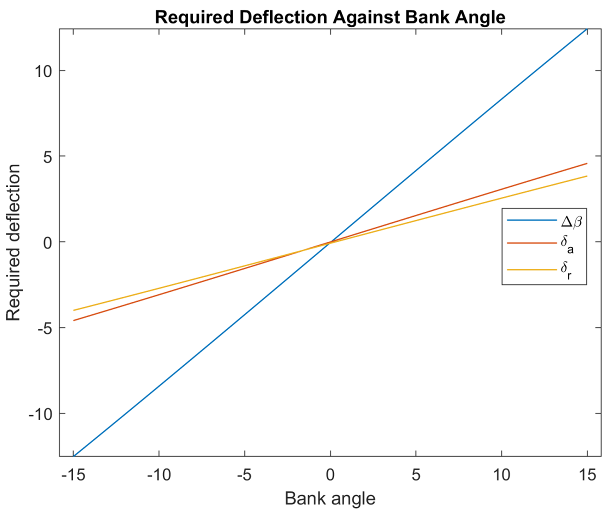

Figure 14: Required Trim Deflection against Bank Angle

deflection angle is well within the maximum aileron and rudder deflection for bank angle within 15^o. It can be concluded that the aircraft is trimmmable during the cruise segment. It is important to note that this linear trend will deviate under large banking maneuvers, as the underlying assumptions no longer hold, causing the required deflections to diverge. Since the trim deflection is not zero at zero bank, this would mean the aircraft have a natural tendency to bank and yaw, this is further reinforced by flight simulation test.

5.4 One Engine Inoperative

In the event of an engine failure, aileron and rudder input will be necessary to compensate the rolling and yawing moment generated from the asymmetric thrust condition. In such case, the moment[9] would entirely due to one propeller.

$$(n)_p = \rho \frac{\omega_p}{2\pi} d_p^5 (C_{n_\alpha})_p (\alpha_{0p} - \epsilon_{d,\alpha}) \tag{60}$$

$$(l)_p = -\frac{P_b}{\omega_p} = -\frac{(TU_\infty/\eta_p)}{\omega_p} \tag{61}$$

Converting $(\Delta n)_p$ and $(\Delta \ell)_p$ to non-dimensional form gives:

$$(\Delta C_n)_p = \frac{n_p - y_{bp}T}{\frac{1}{2}\rho U_\infty^2 S_w b_w} \quad \text{and} \quad (\Delta C_l)_p = \frac{\ell_p}{\frac{1}{2}\rho U_\infty^2 S_w b_w} \tag{62}$$

In trimming the aircraft, we set β to zero, this transform equation (59) to (63).

$$\begin{bmatrix} C_{\ell,\delta_a} & C_{\ell,\delta_r} \\ C_{n,\delta_a} & C_{n,\delta_r} \end{bmatrix} \begin{Bmatrix} \delta_a \\ \delta_r \end{Bmatrix} = \begin{bmatrix} -(\Delta C_\ell)_p \\ -(\Delta C_n)_p \end{bmatrix} \tag{63}$$

$$\delta_a = -C_{\ell,\delta_r}(\Delta C_n)_p - C_{l,\delta_a}(\Delta C_l)_p \tag{64}$$

$$\delta_r = -C_{n,\delta_r}(\Delta C_n)_p - C_{n,\delta_a}(\Delta C_l)_p \tag{65}$$

Required deflection	δ_a	δ_r
Cruise segment 1	-0.9315	-8.9593
Cruise segment 2	-0.8280	-7.9638

Table 12: OEI Required Deflection

Table 12 indicates in the absence of slipstream effects, the aircraft requires approximately 9 degrees of right rudder and slightly less than 1 degree of right aileron for trim. This indicates that ailerons sizing considerations based on OEI at cruise is not significant.

5.5 Minimum Control Airspeed VMCA

The minimum control airspeed is of great interest as it indicates the minimum velocity required to make a bank turn. CS[18] § 25.149 prohibits the minimum control airspeed to be exceed $1.13V_s$.

Expressing[26] C_W in dimensional form, the minimum control airspeed can be computed. Equation (66) gives the VMCA from a fixed rudder deflection $\delta_r \approx \delta_{r_{\max}}$.

$$\begin{bmatrix} C_{l_\beta} & C_{l_{\delta_a}} & 0 \\ C_{n_\beta} & C_{n_{\delta_a}} & \frac{2\Delta T}{\rho S b} \\ C_{Y_\beta} & C_{Y_{\delta_a}} & \frac{2W\sin\phi}{\rho S} \end{bmatrix} \begin{bmatrix} \beta \\ \delta_a \\ 1/U^2 \end{bmatrix} = -\begin{bmatrix} C_{l_{\delta_r}}\delta_r \\ C_{n_{\delta_r}}\delta_r + C_{D_e}\frac{b_p}{b} \\ C_{Y_{\delta_r}}\delta_r \end{bmatrix} \tag{66}$$

Similarly, the VMCA for fixed aileron deflection:

$$\begin{bmatrix} C_{l_\beta} & C_{l_{\delta_r}} & 0 \\ C_{n_\beta} & C_{n_{\delta_r}} & \frac{2\Delta T}{\rho S b} \\ C_{Y_\beta} & C_{Y_{\delta_r}} & \frac{2W\sin\phi}{\rho S} \end{bmatrix} \begin{bmatrix} \beta \\ \delta_a \\ 1/U^2 \end{bmatrix} = -\begin{bmatrix} C_{l_{\delta_a}}\delta_a \\ C_{n_{\delta_a}}\delta_a + C_{D_e}\frac{b_p}{b} \\ C_{Y_{\delta_a}}\delta_a \end{bmatrix} \tag{67}$$

Figure 15 shows the VMCA variation with the aircraft weight. Both the minimum control airspeed are below the stall speed and FAR 25 regulated maximum VMCA throughout the operational range of aircraft weight. This illustrates that full control can be ensured and FAR 25 requirement is met.

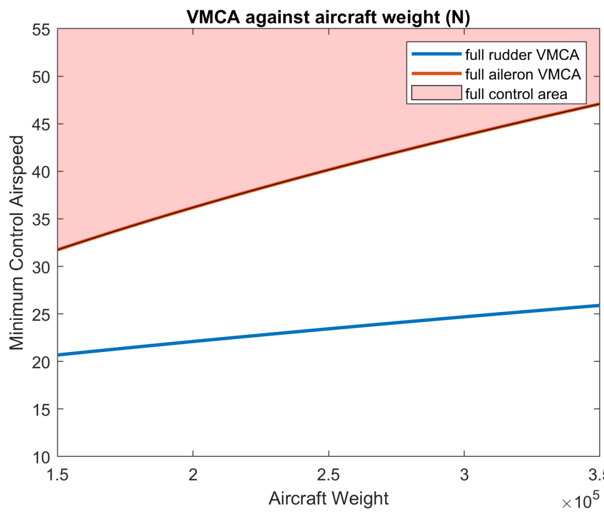

Figure 15: VMCA Variation with Airplane Weight

Table 13 shows the minimum control speed at different flight conditions with margin defined by $1.13V_S - U_{MCA}$ at the particular flight condition. Due to a less constraint environment, cruise segment 2 shows a greater margin.

	VMCA	Margin
Cruise segment 1	39.8	39.3
Cruise segment 2	38.3	40.8
Take off	43.4	15.3
Landing	35.9	14.9

Table 13: VMCA at Different Flight Conditions

5.6 Further Work

Due to time constraints, a comprehensive analysis of lateral stability could not be conducted within the given time frame. In the next iteration, further investigations will focus on considering ground effects and drag, as specified in § 25.149, to meet the necessary requirements. Additionally, according to § 25.149 (d), the rudder force should not exceed 150 pounds during minimum control speed, even when reducing power and thrust of remaining operative engines. To comply with these requirements, it is imperative to examine the forces exerted on both the ailerons and rudder.

In accordance with § 25.149[18], it is mandated that the aircraft's path should not deviate more than 30 feet laterally from the centerline at any point. Given a more flexible time frame, a detailed investigation into the trajectory of the aircraft under OEI conditions will be necessary.

6 Slosh Dynamics and Dynamical Stability

The presence of a free surface in ammonium as a liquid fuel makes it prone to sloshing, causing the fuel inside the tanks to move back and forth during aircraft maneuvers or external disturbances. This sloshing motion can introduce additional dynamics to the system, potentially resonating with the aircraft's dynamic modes and with the control frequency[27], posing challenges for pilot and autopilot control. Consequently, a modal analysis of the fluid field is conducted to determine the natural frequency.

While fuel is stored in the wing box, which has an irregular shape, it can be simplified as a cylindrical tank for ease of analysis. By approximating the cylindrical tank, the fluid field equation can be linearised[28] to account for small displacements and reduce complexity. Solving the Laplace solution of the fluid velocity potential $\tilde{\Phi}_{\text{slosh}}$ yields:

$$\tilde{\Phi}_{\text{slosh}} = \sum_{m=0}^{\infty}\sum_{n=1}^{\infty}[\alpha_{mn}(t)\cos m\theta_{\text{slosh}}+ \beta_{mn}(t)\sin m\theta_{\text{slosh}}]J_m\left(\lambda_{mn}r_{\text{slosh}}\right)\frac{\cosh\left[\lambda_{mn}(z_{\text{slosh}}+h_{\text{slosh}})\right]}{\cosh\lambda_{mn}h_{\text{slosh}}} \tag{68}$$

where $\alpha_{mn}(t)$ and $\beta_{mn}(t)$ are time dependent terms dependent on the free-surface initial conditions. By differentiating once with respect to time and expressing $\alpha_{mn}(t)$ and $\beta_{mn}(t)$ as harmonics $sin(\omega_{mn}t)$, the natural frequency can be expressed as:

$$\omega_{mn}^2 = \frac{g\xi_{mn}}{R}tanh(\xi_{mn}h/R) \tag{69}$$

It can be demonstrated that equation (69) converges to a constant value when the ratio of height to tank radius h/R exceeds 2.

$$\omega_{mn}^2 = \frac{g\xi_{mn}}{R}, \quad \text{for } \xi_{mn}h/R \geq 2.65 \tag{70}$$

At cruise segment, as the fuel has already depleted 30%, this suggests that the height to effective radius h/R is at least 4.5. The aforementioned condition in (70) is met because the first three roots of the asymmetric sloshing modes ξ_{mn}, are 1.841, 5.335, and 8.535, which satisfy the mentioned requirement.

The close proximity between the third natural frequency and the roll subsidence mode could pose a potential issue for the pilot, although further investigation is required to determine the impact of slosh wave amplitude, which was not explored due to time limitations. It should be noted that the assumption of a crude wing box fuel tank and the linearization assumption made in solving the Laplace solution may cause the natural frequency to deviate from the calculated result. In such cases, it becomes necessary to assess the effects of sloshing dynamics on the SPPO and Dutch roll mode to fully understand the implications and potential challenges slosh dynamics poses.

	1	2	3
Slosh natural frequency	3.879	6.6040	8.353

Table 14: First Three Sloshing Natural Frequency

7 Conclusion

The key accomplishments can be summarized as follows. Stability derivatives obtained from computational and approximation methods are utilized to evaluate the stability characteristics of the aircraft. Both the state space model and approximation approaches are employed to analyze the aircraft's dynamic modes using these stability derivatives. Lastly, the lateral stability is assessed to determine compliance with the regulations in § 25.149.

7.1 Further Work

The techniques employed in this report primarily involve analytical methods, which would benefit from verification and validation through computational data. The Vortex Lattice Method is utilized, given its low fidelity and simplicity, particularly during the preliminary design phase. However, given additional time, the use of computational fluid dynamics techniques would provide greater accuracy[29], leading to a more detailed stability and control analysis. Additionally, the inclusion of wind tunnel testing and a scaled model would offer a more comprehensive understanding of the aircraft's behavior in real-life scenarios.

As stated in Section 5.6 of this report, it should be noted that a comprehensive analysis regarding the requirements outlined in CFR § 25.149 could not be fully conducted due to certain limitations, including time constraints and data availability. While the analysis performed thus far provides valuable insights into the aircraft's stability and compliance, there are aspects that remain unexplored. To ensure a more robust assessment and to obtain a more accurate understanding of the aircraft's compliance with the regulatory requirements, further work is highly recommended.

7.2 Summary

To conclude, the analysis conducted confirms that the aircraft shows favorable dynamic and lateral stability as well as longitudinal stability[30]. It meets the regulatory criteria of achieving a 5-degree bank angle within the specified mission profile range, as outlined in CFR § 25.149. The assessment of dynamical modes using a linearized state space model indicates a generally stable dynamic response. While there are minor disparities between the state space model solution and flight simulation data, there is a consistent trend and similarity in the observed aircraft behavior.

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9 Appendix

Equation of κ_β

$$\kappa_\beta(\bar{x},\bar{y},0) = \frac{4\bar{y}(\bar{x}-\kappa_b\tan\Lambda)\kappa_b^2}{\pi^2\left(\bar{y}^2+\kappa_b^2\right)^2}\left[1+\frac{\bar{x}-\kappa_b\tan\Lambda}{\sqrt{(\bar{x}-\kappa_b\tan\Lambda)^2+\bar{y}^2+\kappa_b^2}}\right] + \frac{2\bar{y}\kappa_b}{\pi^2\left(\bar{y}^2+\kappa_b^2\right)}\left[\frac{(\bar{x}-\kappa_b\tan\Lambda)^2\kappa_b}{\left[(\bar{x}-\kappa_b\tan\Lambda)^2+\bar{y}^2+\kappa_b^2\right]^{3/2}}\right]$$

Conversion of lateral derivatives

Y	l	n
$Y_\beta = \frac{qS}{m}C_{y_\beta}$	$l_\beta = \frac{qSb}{I_{xx}}C_{l_\beta}$	$N_\beta = \frac{qSb}{I_{zz}}C_{n_\beta}$
$Y_p = \frac{qS}{m}C_{y_p}$	$l_p = \frac{qSb}{I_{xx}}C_{l_p}$	$N_p = \frac{qSb}{I_{zz}}C_{n_p}$
$Y_r = \frac{qS}{m}C_{y_r}$	$l_r = \frac{qSb}{I_{xx}}C_{l_r}$	$N_r = \frac{qSb}{I_{zz}}C_{n_r}$
$Y_{\delta a} = \frac{qS}{m}C_{y_{\delta a}}$	$l_{\delta,a} = \frac{qSb}{I_{xx}}C_{l_{\delta a}}$	$N_{\delta a} = \frac{qSb}{I_{zz}}C_{n_{\delta a}}$
$Y_{\delta,r} = \frac{qS}{m}C_{y_{\delta r}}$	$l_{\delta r} = \frac{qSb}{I_{xx}}C_{l_{\delta r}}$	$N_{\delta r} = \frac{qSb}{I_{zz}}C_{n_{\delta r}}$